

Findings-first economics

# Acting on these findings is modelled to improve health and reduce healthcare spending.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

INCREMENTAL QALYS

**0.071**

about 26 days of healthy life, on average, for a person with this pattern of findings

INCREMENTAL HEALTHCARE COST

**-\$26,549**

healthcare-sector perspective; negative means the modelled programme saves money

REFERENCE-CASE NMB

**\$33,643**

deterministic, at \$100,000 per QALY — health valued at that threshold, not cash

DECISION UNCERTAINTY

**1,500 of 1,500**

modelled simulations cost-effective (100%); cost-saving in all 1,500 of 1,500 simulations (100%) · 95% interval \$15,886 to \$59,856 across 62 varied parameters

22

**findings carry a standardised economic estimate**

6 high-confidence, 9 moderate-confidence, 8 low-confidence, 30 ungraded. 31 of the 53 findings shown are reported without a dollar figure rather than being assigned one the model cannot support.

BRCA1 pathogenic variant

LDLR familial hypercholesterolemia

APOB familial hypercholesterolemia

APOE ε4 carrier

PTPN22 R620W carrier

CHEK2 I157T carrier

Factor V Leiden / Prothrombin...

HLA-DQ2/DQ8 carrier

HBB carrier

SERPINA1 carrier

OPRM1 OPRM1 A118G (mu-opioid...

LRRK2 G2019S carrier

CYP2C19 Intermediate Metabolizer

G6PD deficiency

DRD2/ANKK1 DRD2 / ANKK1 Taq1A

CYP2D6 Intermediate Metabolizer

NAT2 — NAT2 slow acetylator —...

PON1 — PON1 Q192R —...

SLCO1B1 Intermediate Function

ATP7B variant

Coeliac-permissive HLA haplotype...

Vitamin D Synthesis Efficiency

and 31 more

## What rises to the top

Ranked by expected net monetary benefit, each valued on its own.

Finding-level values are not additive; the pooled reference case is calculated separately.

BRCA1 PATHOGENIC VARIANT

**\$18,808**

Enhanced screening + risk-reducing surgery option

HIGH

LDLR FAMILIAL HYPERCHOLESTEROLEMIA

**\$9,442**

High-intensity statin + cascade screening

HIGH

APOB FAMILIAL HYPERCHOLESTEROLEMIA

**\$6,658**

High-intensity statin therapy

HIGH

APOE E4 CARRIER

**\$3,514**

Intensive cardiovascular risk reduction (statin) to prevent MI...

MODERATE

Your findings at a glance

Findings that could change a decision

Every finding the model examined, the decision it bears on, the strength of the evidence, and what it is worth on its own. No dollar figure appears without the genomic result that produced it — the detail behind each figure is on the pages that follow.

MEDICATION & PRESCRIBING 3 findings

|                                                                  |                         |      |       |
|------------------------------------------------------------------|-------------------------|------|-------|
| CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)              | Clopidogrel prescribing | HIGH | \$490 |
| CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants) | Codeine prescribing     | HIGH | \$210 |
| SLCO1B1 Intermediate Function (IM-like) (simvastatin)            | Simvastatin prescribing | HIGH | \$90  |

RISK & PREVENTION 12 findings

|                                                                     |                                               |          |          |
|---------------------------------------------------------------------|-----------------------------------------------|----------|----------|
| BRCA1 pathogenic variant (heterozygous)                             | Enhanced screening + risk-reducing surgery... | HIGH     | \$18,808 |
| LDLR familial hypercholesterolemia (heterozygous)                   | High-intensity statin + cascade screening     | HIGH     | \$9,442  |
| APOB familial hypercholesterolemia (heterozygous)                   | High-intensity statin therapy                 | HIGH     | \$6,658  |
| APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)        | Intensive cardiovascular risk reduction...    | MODERATE | \$3,514  |
| PTPN22 R620W carrier — autoimmune screening awareness               | Awareness for autoimmune symptom clusters     | MODERATE | \$2,509  |
| CHEK2 I157T carrier — moderate cancer risk awareness                | Risk-aware screening schedule                 | MODERATE | \$2,449  |
| Factor V Leiden / Prothrombin carrier — situational anticoagulation | Awareness for high-risk situations            | MODERATE | \$2,273  |
| HLA-DQ2/DQ8 carrier — celiac rule-out capability                    | Negative DQ2/DQ8 rules out celiac             | MODERATE | \$2,097  |
| HBB carrier (reproductive)                                          | —                                             | MODERATE | \$1,226  |

# Findings that could change a decision, continued

RISK & PREVENTION (CONTINUED) 12 findings

|                                                                      |                                                  |          |                      |
|----------------------------------------------------------------------|--------------------------------------------------|----------|----------------------|
| SERPINA1 carrier (reproductive)                                      | —                                                | MODERATE | \$1,226              |
| G6PD deficiency — oxidative drug avoidance                           | Avoid primaquine/dapsone/rasburicase...          | MODERATE | \$463                |
| Coeliac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision     | tTG-IgA serology if symptomatic, before any...   | MODERATE | -\$122               |
| LOWER-CONFIDENCE & EXPLORATORY 8 findings                            |                                                  |          |                      |
| OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G...              | OPRM1-guided analgesic dosing + AUD treatment... | LOW      | \$699                |
| LRRK2 G2019S carrier — Parkinson's early intervention                | Prodromal PD monitoring + neuroprotective...     | LOW      | \$613                |
| DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway... | Dopamine-aware addiction treatment +...          | LOW      | \$442                |
| NAT2 — NAT2 slow acetylator — occupational cancer risk awareness     | Aromatic amine exposure monitoring + bladder...  | LOW      | \$90                 |
| PON1 — PON1 Q192R — organophosphate susceptibility                   | Pesticide exposure avoidance + cholinesterase... | LOW      | \$90                 |
| ATP7B variant — Wilson's disease copper monitoring                   | Serum copper/ceruloplasmin monitoring + early... | LOW      | -\$9                 |
| Vitamin D Synthesis Efficiency — Nutrient-metabolism variant...      | Targeted supplementation                         | LOW      | -\$216               |
| 2 actionable wellness variant(s)                                     | Nutrient metabolism, sleep, and fitness...       | LOW      | Not yet standardised |

AWARENESS — NOT MONETISED 30 findings

|                                                             |                                   |          |               |
|-------------------------------------------------------------|-----------------------------------|----------|---------------|
| Avoid primaquine/dapsone/rasburicase hemolytic crisis       | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| OPRM1-guided analgesic dosing + AUD treatment matching      | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Dopamine-aware addiction treatment + antipsychotic response | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Aromatic amine exposure monitoring + bladder screening      | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Pesticide exposure avoidance + cholinesterase monitoring    | Unvaluable — incomplete economics | NO GRADE | Not monetised |

# Findings that could change a decision, continued

AWARENESS — NOT MONETISED (CONTINUED) 30 findings

|                                                                       |                                   |          |               |
|-----------------------------------------------------------------------|-----------------------------------|----------|---------------|
| Awareness for autoimmune symptom clusters                             | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Targeted supplementation (vitamin D/B12/folate/iron)                  | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Awareness for high-risk situations (surgery, pregnancy, long flights) | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Serum copper/ceruloplasmin monitoring + early chelation               | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Risk-aware screening schedule                                         | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Partner testing for reproductive risk assessment                      | Carrier Screening                 | NO GRADE | Not monetised |
| Intensive cardiovascular risk reduction (statin) to prevent MI...     | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| High-intensity statin + cascade screening                             | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| High-intensity statin therapy                                         | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Enhanced screening + risk-reducing surgery option                     | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Negative DQ2/DQ8 rules out celiac                                     | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Avoid clopidogrel non-response / stent thrombosis (MACE)              | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Avoid opioid toxicity or ineffective analgesia/SSRI dosing            | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| tTG-IgA serology if symptomatic, before any gluten withdrawal...      | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Prevent statin-induced myopathy / rhabdomyolysis                      | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Prodromal PD monitoring + neuroprotective trial eligibility           | Unvaluable — incomplete economics | NO GRADE | Not monetised |

# Findings that could change a decision, continued

AWARENESS — NOT MONETISED (CONTINUED) 30 findings

|                                                                          |                                   |          |               |
|--------------------------------------------------------------------------|-----------------------------------|----------|---------------|
| Personalized AUD risk awareness + naltrexone/acamprosate matching        | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| OPRM1-guided opioid dosing + naltrexone response prediction              | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| CHRNA5-guided smoking cessation (varenicline dosing)                     | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Trauma-informed care + HPA-axis-targeted therapy selection               | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Avoid clopidogrel non-response / stent thrombosis (MACE)                 | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Avoid opioid toxicity or ineffective analgesia/SSRI dosing               | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Prevent statin-induced myopathy / rhabdomyolysis                         | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Prevent major bleeding event from warfarin overdosing                    | Unvaluable — incomplete economics | NO GRADE | Not monetised |
| Reproductive genetics — 12 carrier condition(s) where partner testing... | Family Planning                   | NO GRADE | Not monetised |

**Finding-level values are standalone expected values. They are not additive.** The reference-case total is calculated separately after overlapping signals are pooled by condition.

Expected NMB values health at \$100,000 per QALY and is **not cash returned to anyone**. Ordering is by clinical group first and expected NMB second.

31 findings are shown without a standardised expected NMB rather than being assigned a value the model cannot support.

Medication-genotype findings

# Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

### CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)

clopidogrel

**WHAT WAS FOUND**

Genotype predicts altered handling of clopidogrel.

**WHY IT MATTERS**

Clopidogrel is a prodrug that CYP2C19 must activate. Reduced activity means less active drug and a higher chance the antiplatelet effect is inadequate.

**WHAT COULD CHANGE**

Avoid clopidogrel non-response / stent thrombosis (MACE)

**KEY DEPENDENCY**

Value accrues only if clopidogrel is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

**\$490**

|                      |            |
|----------------------|------------|
| Medical cost averted | \$300      |
| Health gain          | 0.002 QALY |
| Cost to act          | \$10       |

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

### CYP2D6 Intermediate Metabolizer (IM) (codeine /...

codeine antidepressants

**WHAT WAS FOUND**

Genotype predicts altered handling of codeine antidepressants.

**WHY IT MATTERS**

CYP2D6 activity governs conversion of codeine to morphine and clearance of several antidepressants, so one dose can be ineffective in some people and excessive in others.

**WHAT COULD CHANGE**

Avoid opioid toxicity or ineffective analgesia/SSRI dosing

**KEY DEPENDENCY**

Value accrues only if codeine antidepressants is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

**\$210**

|                      |            |
|----------------------|------------|
| Medical cost averted | \$120      |
| Health gain          | 0.001 QALY |
| Cost to act          | \$30       |

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

### SLCO1B1 Intermediate Function (IM-like) (simvastatin)

simvastatin

**WHAT WAS FOUND**

Genotype predicts altered handling of simvastatin.

**WHY IT MATTERS**

SLCO1B1 transports statins into the liver. Reduced function raises systemic exposure and with it the risk of muscle toxicity, most clearly for simvastatin.

**WHAT COULD CHANGE**

Prevent statin-induced myopathy / rhabdomyolysis

**KEY DEPENDENCY**

Value accrues only if simvastatin is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

**\$90**

|                      |            |
|----------------------|------------|
| Medical cost averted | \$60       |
| Health gain          | 0.001 QALY |
| Cost to act          | \$20       |

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on the methods page.

**Decision support, not a directive.** Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk, prevention & exploratory findings

Findings with weaker evidence, reported anyway

Lower confidence changes how a finding should be read, not whether it is shown. A negative expected NMB means the modelled cost of acting exceeds the modelled benefit — useful information, not an error. Evidence grade reflects the strength of the association, not the size of the value; provenance is on the methods page.

|                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <div><div>BRCA1 pathogenic variant (heterozygous)</div><div>POTENTIAL DECISION</div><div>Enhanced screening + risk-reducing surgery option</div></div> <div><div>EXPECTED NMB</div><div>\$18,808</div><div>medical cost averted \$17,412</div><div>health gain 0.019 QALY</div><div>cost to act \$500</div></div>                                                                               | HIGH     |
| <div><div>LDLR familial hypercholesterolemia (heterozygous)</div><div>POTENTIAL DECISION</div><div>High-intensity statin + cascade screening</div></div> <div><div>EXPECTED NMB</div><div>\$9,442</div><div>medical cost averted \$9,136</div><div>health gain 0.008 QALY</div><div>cost to act \$500</div></div>                                                                               | HIGH     |
| <div><div>APOB familial hypercholesterolemia (heterozygous)</div><div>POTENTIAL DECISION</div><div>High-intensity statin therapy</div></div> <div><div>EXPECTED NMB</div><div>\$6,658</div><div>medical cost averted \$6,578</div><div>health gain 0.006 QALY</div><div>cost to act \$500</div></div>                                                                                           | HIGH     |
| <div><div>APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)</div><div>POTENTIAL DECISION</div><div>Intensive cardiovascular risk reduction (statin) to prevent MI — Intensive statin therapy + lipid monitoring</div></div> <div><div>EXPECTED NMB</div><div>\$3,514</div><div>medical cost averted \$3,869</div><div>health gain 0.001 QALY</div><div>cost to act \$500</div></div> | MODERATE |
| <div><div>PTPN22 R620W carrier — autoimmune screening awareness</div><div>POTENTIAL DECISION</div><div>Awareness for autoimmune symptom clusters</div></div> <div><div>EXPECTED NMB</div><div>\$2,509</div><div>medical cost averted \$2,579</div><div>health gain 0.004 QALY</div><div>cost to act \$500</div></div>                                                                           | MODERATE |

Where no finding-specific caveat is shown: baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis, and neither is measured for this individual.

**Reported without a standardised value (31):** 2 actionable wellness variant(s), Avoid primaquine/dapsone/rasburicase..., OPRM1-guided analgesic dosing + AUD..., Dopamine-aware addiction treatment +..., Aromatic amine exposure monitoring +..., Pesticide exposure avoidance +..., and 25 more. No registry-backed economic pathway exists for these yet, so none is invented. Each appears with its decision and evidence grade on the glance pages.



How the findings combine

Taken together, what do these findings mean?

This is the canonical reference case. Findings bearing on the same condition are combined on the risk scale, each condition's cost of illness is charged exactly once, and trial efficacy is discounted to real-world adherence before any value is credited.

INCREMENTAL  
HEALTHCARE COST

**-\$26,549**

healthcare-sector perspective

INCREMENTAL QALYS

**0.071**

about 26 days of healthy life on average

REFERENCE-CASE NMB

**\$33,643**

at \$100,000 per QALY

CLASSIFICATION

**dominant (more health, lower cost)**

not defined — strategy dominates or is dominated

Condition pools — combined before value is credited

| CONDITION     | FINDINGS | BASELINE | RAW<br>UNCORRECTED | POOLED | ADHERENCE | REALISED | COST AVERTED | QALYS  | NMB      |
|---------------|----------|----------|--------------------|--------|-----------|----------|--------------|--------|----------|
| BreastOvarian | 1        | 60.0%    | 45%                | 45%    | 65%       | 29%      | \$13,625     | 0.015  | \$14,784 |
| CAD           | 3        | 50.0%    | 98%                | 60%    | 50%       | 34%      | \$7,410      | 0.022  | \$9,339  |
| Pathogenic    | 6        | 20.0%    | 220%               | 60%    | 65%       | 48%      | \$5,019      | 0.017  | \$6,368  |
| Alzheimer     | 1        | 15.0%    | 30%                | 30%    | 35%       | 11%      | \$1,630      | 0.001  | \$1,516  |
| IronOverload  | 3        | 20.0%    | 32%                | 17%    | 65%       | 12%      | \$478        | 0.010  | \$1,148  |
| Depression    | 2        | 20.0%    | 18%                | 13%    | 50%       | 7%       | \$379        | 0.007  | \$817    |
| Coeliac       | 1        | 20.0%    | 30%                | 30%    | 65%       | 20%      | \$283        | 0.0001 | -\$29    |

Raw is an uncorrected sum, not a modelled result. It adds the risk reductions of findings that share a condition, which is not how probabilities combine. Any value above 100% is the artifact this correction removes. The pooled figure beside it is the model's answer.

Two corrections, applied in order

The engine reports these as one combined reduction. They rest on different evidence and are separated here: overlap is a structural correction, adherence a behavioural one.

NAIVE AVOIDED COST

**\$60,291**

every finding counted independently

POOLED, AT TRIAL EFFICACY

**\$46,930**

22% removed — one liability, one cost of illness

REALISED BENEFIT

**\$28,824**

39% further removed — efficacy is not effectiveness

Adherence is charged to the benefit *and* to ongoing intervention cost, but not to the one-off test cost. 0.042 QALYs are lost to imperfect adherence relative to trial efficacy.

**\$33,643**

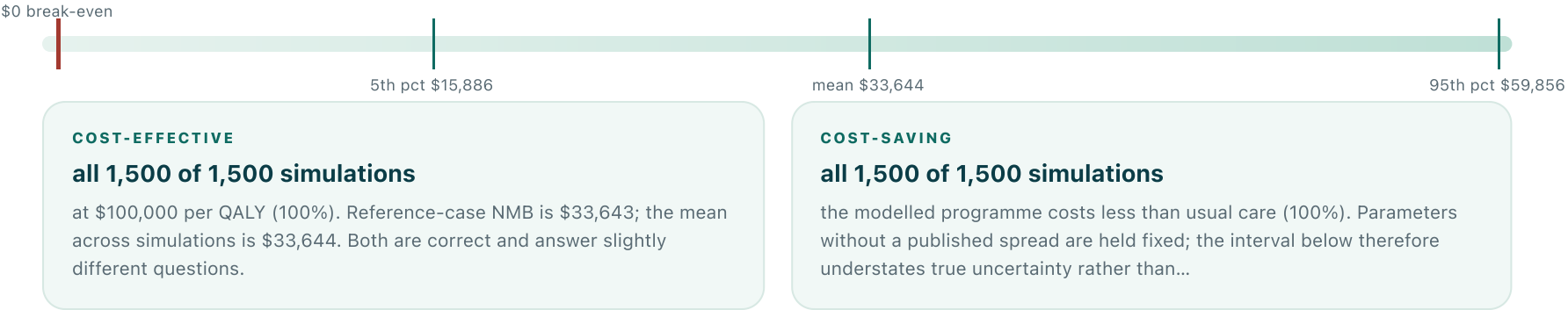


Uncertainty, evidence & model discipline

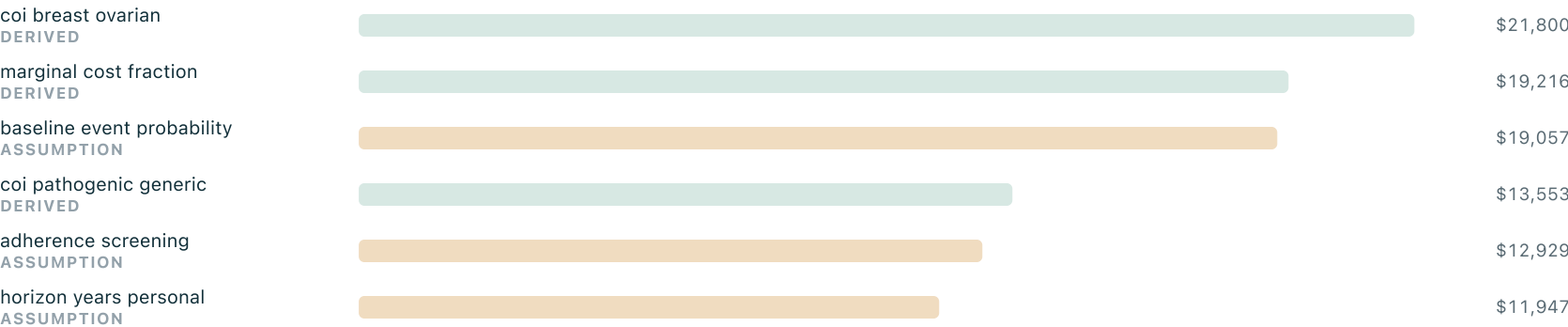
# How sure is this, and what rests on judgement?

Uncertainty has two layers: the spread of modelled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

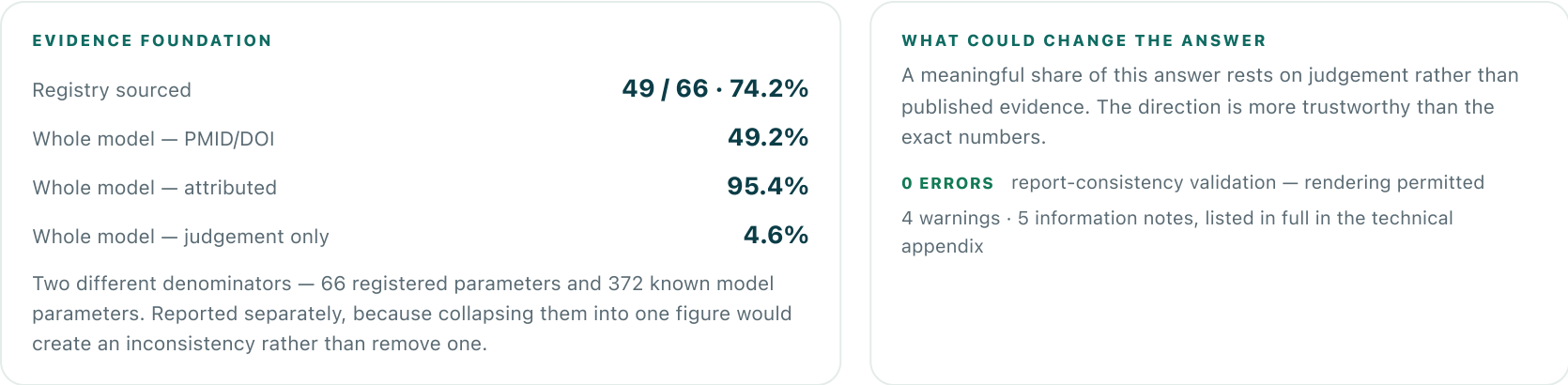
## Net monetary benefit across 1,500 simulations



## What drives the answer



Swing in net monetary benefit across each parameter's plausible range. Amber bars are declared assumptions rather than sourced values; they account for 39.7% of total swing.



The testing decision

Is whole-genome sequencing worth it?

Two different questions live on this page and they are never added together. One is a choice made *before* testing, averaged over a population. The other counts what sequencing actually found in this genome.

**PROSPECTIVE — BEFORE TESTING**

**What is the expected value of choosing sequencing rather than an array?**

|                                              |                 |
|----------------------------------------------|-----------------|
| Chance an array misses an actionable finding | <b>1 in 56</b>  |
| Value if you are that person                 | <b>\$25,189</b> |
| Expected value, averaged over everyone       | <b>\$491</b>    |
| Extra cost over an array                     | <b>\$200</b>    |
| Net expected value                           | <b>\$291</b>    |

**OBSERVED — IN THIS GENOME**

**What did sequencing actually find that an array could not?**

|                                  |                |
|----------------------------------|----------------|
| Sequencing-only findings present | <b>2</b>       |
| Observed standalone contribution | <b>\$2,452</b> |

A count of findings this genome contains, not an expectation. For this input the population question is already settled.

**Not the same as the value of sequencing.** This line counts only findings an array *cannot* report. Sequencing also recovers findings an array could have carried and did not, which is the larger effect — so the full chip-to-whole-genome difference is bigger than the figure above.

**Separation rule.** The prospective figures describe the population this genome was drawn from; the observed figure describes the genome itself. They answer different decision questions, come from different inputs, and are never combined into one headline total.

Strategy frontier

| STRATEGY                | COST  | QALYS  | ICER VS PREVIOUS | STATUS    |
|-------------------------|-------|--------|------------------|-----------|
| No testing              | \$0   | 20.000 | —                | efficient |
| Genotyping chip         | \$200 | 20.060 | \$3,333/QALY     | efficient |
| Whole-genome sequencing | \$600 | 20.120 | \$6,667/QALY     | efficient |
| Targeted clinical panel | \$900 | 20.090 | —                | dominated |

Frontier values illustrate the testing choice for a population and are not this genome's reference case. This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right...

Methods & provenance

# What was assumed, where it came from, what was checked

The pages before this one tell the story; this page is their audit trail. Full parameter tables, references and the advanced analyses follow in the technical appendix.

| MODEL              |                          | PROVENANCE                                                                                              |       |
|--------------------|--------------------------|---------------------------------------------------------------------------------------------------------|-------|
| Perspective        | healthcare sector        | Registered parameters                                                                                   | 66    |
| Comparator         | usual care               | Published                                                                                               | 8     |
| Outcomes           | cost, QALYs, NMB         | Derived                                                                                                 | 41    |
| Time horizon       | 10 years                 | Declared assumptions                                                                                    | 17    |
| Discounting        | 3%, costs and QALYs      | Registry sourced                                                                                        | 74.2% |
| Willingness to pay | \$100,000/QALY           | Model, resolvable                                                                                       | 49.2% |
| Currency           | USD, United States       | Model, attributed                                                                                       | 95.4% |
| Pooling            | complement of products   | Model, judgement only                                                                                   | 4.6%  |
| Adherence          | benefit and ongoing cost | A parameter declared as an assumption may not cite a source — the registry refuses to load if one does. |       |
| Uncertainty        | 1,500-run PSA            |                                                                                                         |       |

## Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labelled as such.
- Costs are not inflation-normalised to a single price year.
- Findings without a registry-backed pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

## Validation

- PASS** NMB identity —  $\lambda \times \Delta QALY - \Delta Cost$  reconciles to the reported total

**PASS** Condition cost charged once — no pool exceeds its cost of illness

**PASS** WGS basis separation — observed and prospective never combined

**PASS** Report validation gate — no errors; rendering permitted
- PASS** Dominance classification — sign of cost and effect agrees with the label

**PASS** Pooled risk cap — combined risk reduction stays within its bound

**PASS** Budget-impact consistency — peak burden is the maximum net spend

**Audit trail.** Every figure in this report is serialised in `economics-payload.json`, the canonical object these pages render from. The parameter registry, references, cross-path reconciliation and the advanced analyses follow in the technical appendix. Build GENOMELENS-BUILD: 41d04f5+dirty 2026-09-05T00:03:15-04:00. Synthetic input — no human genome was used.